

6. PRIMARY AND SECONDARY RESEARCH: METHODS, COSTS, ACCESS, AND ETHICS

This section should include evidence of the use of appropriate methods of primary and secondary research including consideration of cost, access, and ethical issues. This section will cover all research work carried out for this computing project.

6.1 PRIMARY RESEARCH SURVEY DESIGN AND EXECUTION

The main research tool was a 15-question survey that was circulated through Google Forms for 21 days (April–May 2026). The survey targeted Critical Infrastructure (CI) and cybersecurity professionals from eight sectors. A purposive sampling method was used to guarantee relevant expertise. The sample obtained was $n=22$ valid responses across eight sectors such as Finance/Banking, Energy, Healthcare and Defence.

Cost: Google Forms is a free service. No incentives were provided for participation. Cost of primary data collection: £0. This allowed the survey to be financially feasible in an academic research setting.

Access: Participants were recruited through LinkedIn professional networks, IISP (Institute of Information Security Professionals) forums, and direct contacts. Postings of recruitment in specialist CI security groups. The 21-day collection period allowed adequate response time for various time zones. Institutional gatekeeper approval was not required other than TUF ethics oversight.

Ethical Considerations: Complete informed consent was acquired electronically prior to participation. No identifying information was collected and all responses were anonymized before analysis. Data was stored on UK based servers in accordance with UK GDPR. All participants were told they could withdraw at any time without penalty.

6.2 SECONDARY RESEARCH SYSTEMATIC LITERATURE REVIEW

The secondary research has been done by systematic literature review from IEEE Xplore, ACM Digital Library, Google Scholar, and NCSC publications. The following inclusion criteria were applied to filter the sources: published within 5 years, peer-reviewed or of institutional authority, and directly related to at least one research question. Ten sources were selected that met the criteria.

Cost : University of Faisalabad TUF International Programmers is subscribed to IEEE Xplore and ACM Digital Library. Google Scholar and NCSC publications are available without any charge. Total cost of secondary research: £0.

Access: 10/10 sources were accessible through institutional or open access channels. No paywalled sources had to be individually purchased. NCSC guidance documents can be found at gov.uk. This provided complete replicability of the literature review.

Ethical Considerations: All sources have been properly referenced using the Harvard referencing style. No copyrighted material was used beyond the limits of fair use citation. All referenced

authors' intellectual property rights are respected. There were no primary findings reported from the data in commercial reports.

6.3 ALGORITHM SIMULATIONS
SWAT AND CIC-IDS DATASETS

Two publicly available data sets were used for algorithm benchmarking: the Secure Water Treatment (SWaT) dataset from Singapore University of Technology and Design and the CIC-IDS (Canadian Institute for Cybersecurity Intrusion Detection) dataset. Both are cited extensively in literature and can be used for OT security research.

Cost: SWaT and CIC-IDS datasets can be used for free for academic research. A containerized air-gapped environment was used for simulation. Cost of Infrastructure for simulation and adversarial testing: ~ £300 (Compute time, Storage, Setup of containment environment). This is a proportionate cost for a research project of this size.

Access: Both datasets were accessed via institutional academic registration. The air-gapped simulation environment maintained that no live CI systems were accessed, tested or affected at any point.

Ethical Considerations: All adversarial payloads were confined in the air-gapped environment as per the NCSC responsible disclosure principles. No live Critical Infrastructure was tested, accessed, or put at risk. All tests were conducted on mock data. Ethical approval was provided in Appendix C.

6.4 RESEARCH METHODS SUMMARY

Method	Type	Cost	Access Route	Ethical Consideration
Survey (n=22)	Primary / Quantitative	£0	LinkedIn, IISP, direct	Anonymized; GDPR; informed consent
Literature Review	Secondary	£0 (TUF access)	IEEE, ACM, Google Scholar, NCSC	Harvard attribution; fair use

Method	Type	Cost	Access Route	Ethical Consideration
Algorithm Simulation	Primary / Experimental	~£220	SWaT/CIC-IDS academic datasets	Air-gapped; NCSC disclosure principles
Adversarial Testing	Primary / Experimental	~£80	Air-gapped digital twin	Payloads contained; no live CI affected